Chapter 3
Introduction to Regression

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MGMTMSA 400

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Conditional Prediction

Conditional Prediction

- **Conditional prediction**: use information about X to predict Y
  - Y: the outcome we want to predict, also called the response, dependent, or **output** variable
  - X: the variable we use to make the prediction, also called the predictor, explanatory, independent, or **input** variable
- A **prediction rule** maps an input to a predicted outcome: $\hat{Y} = f(X)$
  - In our data, Y denotes the observed outcome, while $\hat{Y}$ denotes a predicted (not observed) value

Example: Predicting Market Prices for Video Cards

- **Problem**
  - Suppose a manufacturer plans to launch a video card with a given performance score. What price would we predict based on similar products already on the market?
- **Solution**
  - Look at existing video cards for which we know the price and performance score
  - Build a **prediction rule** that predicts price as a function of performance score
- **Roadmap for Chapter 3**
  - How do we use data to construct a prediction rule $f(X)$?
  - How do we measure how well the prediction rule fits the observed data?
  - (In Chapter 4) How do we measure uncertainty when predicting an unobserved outcome for a given X?

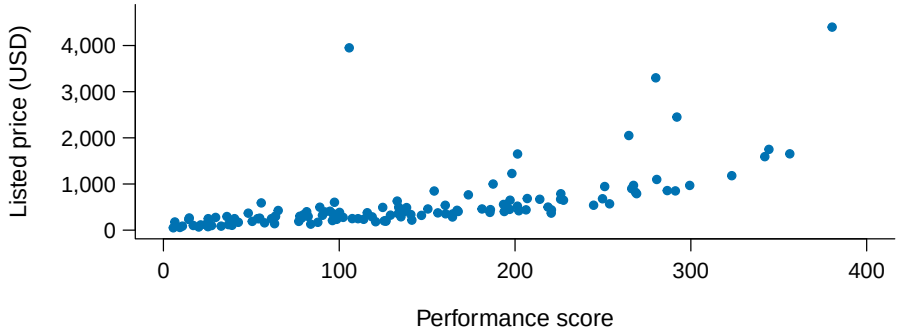
Video Card Pricing Data

- `videocards.csv` contains listed prices and performance scores for 125 video cards
  - Prices and scores were collected from PassMark on October 2, 2026
  - See `videocards_data_dictionary.txt` for the variable definitions
- G3D Mark summarizes performance across several graphics tests; a higher score indicates better performance on this benchmark

```
vc <- read.csv("../../Data/videocards.csv")
head(vc[, 1:6])
##   gpu_id              model price_usd g3d_mark category_as_reported g2d_mark
## 1   3757    GeForce GT 1030    119.99     2395              Desktop      446
## 2   2177 GeForce GTX 650 Ti     75.36     2536              Desktop      421
## 3   2152    GeForce GTX 660    249.00     4044              Desktop      493
## 4   1458   GeForce GTX 660M    229.99     1443               Mobile      275
## 5   2825    GeForce GTX 750     86.99     3283              Desktop      521
## 6   2815 GeForce GTX 750 Ti    104.99     3900              Desktop      534
vc$performance <- vc$g3d_mark / 100
```

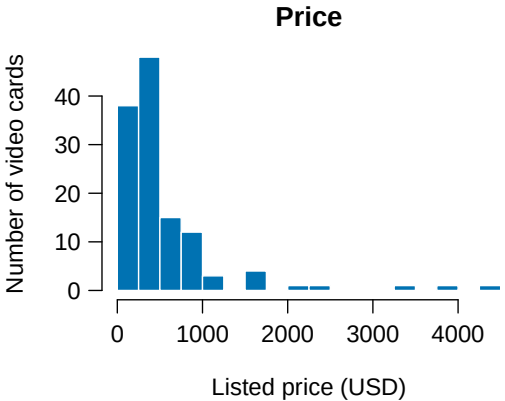
Video Card Pricing Data: Outlier

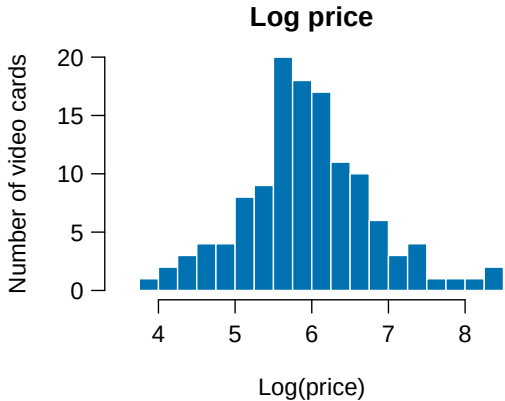
- A **scatter plot** shows the relationship between two variables; each point here represents one video card
- An **outlier** has a Y value that is unusual compared with observations with similar values of X
- Which points look unusual? Would you remove any of them? Why?



Video Card Pricing Data: Log Transformation

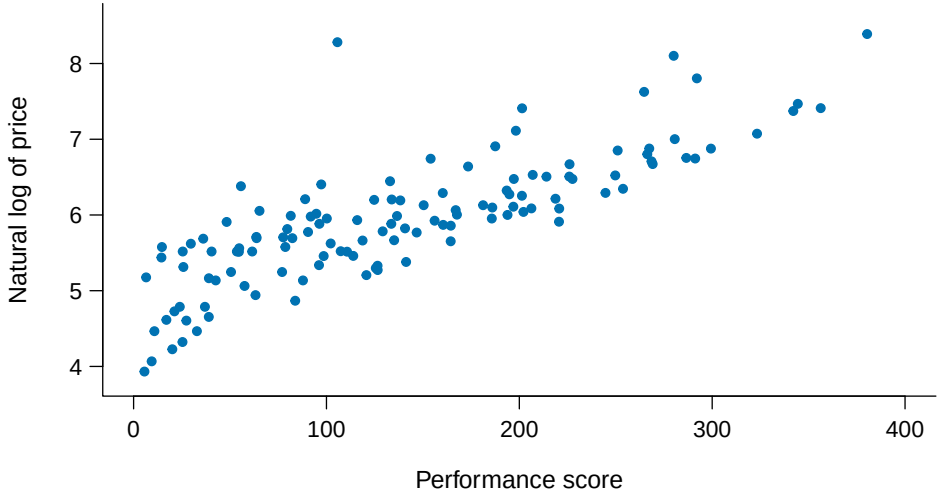
- Prices have a long right tail: a few video cards cost much more than the others
- Taking logs compresses high prices and makes the distribution less skewed





Video Card Pricing Data: Log Price and Performance

- Revisit the scatter plot, but between performance scores and logged prices



Linear Fitting and Prediction

Linear Fitting

- A linear prediction rule has the form

$$\hat{Y} = b_0 + b_1 X$$

  - b_0 is the intercept: the predicted value of Y when $X = 0$
  - b_1 is the slope: the change in the predicted value of Y when X increases by one unit

- In our video card example, X is the performance score, and Y is the natural log of the listed price in dollars

Linear Fitting: Results

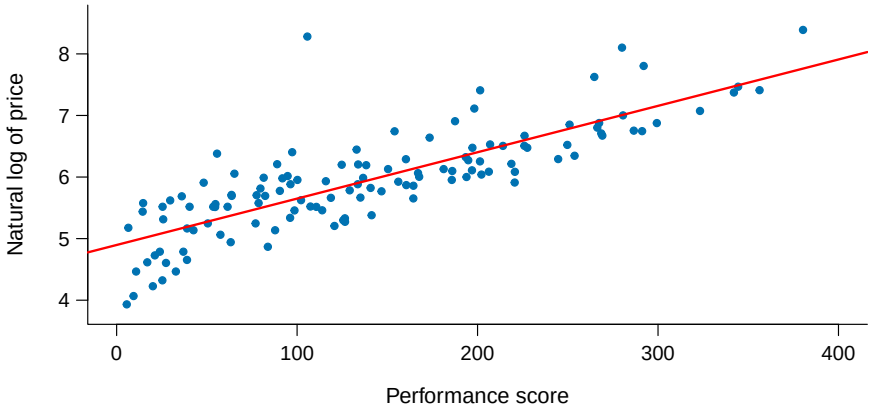
- Fit a line using the R function `lm()`
- The fitted line is $\hat{Y} = 4.897 + 0.008 \cdot X$

```
outlm <- lm(log(price_usd) ~ performance, data = vc)
summary(outlm)
##
## Call:
## lm(formula = log(price_usd) ~ performance, data = vc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.00693 -0.29504 -0.09091  0.26087  2.58868
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4.8968245  0.0800635   61.16   <2e-16 ***
## performance 0.0075311  0.0004826   15.61   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4855 on 123 degrees of freedom
## Multiple R-squared:  0.6644, Adjusted R-squared:  0.6617
## F-statistic: 243.6 on 1 and 123 DF,  p-value: < 2.2e-16
```

Linear Prediction

Add the fitted line to the scatter plot using `abline()`

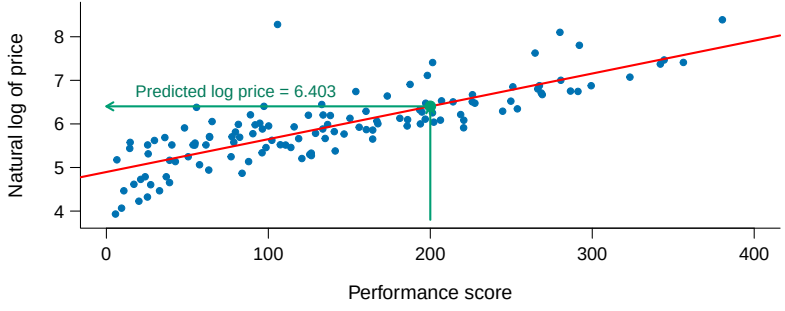
```
abline(outlm, lwd = 2, col = "red")
```



Linear Fitting and Prediction in R

Predict the price of a video card with a performance score of 200

```
log_price_hat <- predict(outlm, newdata = data.frame(performance = 200))
cat("Predicted log price:", round(log_price_hat, 3), "\n")
## Predicted log price: 6.403
cat("Predicted price (USD):", round(exp(log_price_hat), 2), "\n")
## Predicted price (USD): 603.68
```



- `predict()` gives a log price; `exp()` converts it into a price prediction of about $604

Linear Fitting and Prediction: Key Questions

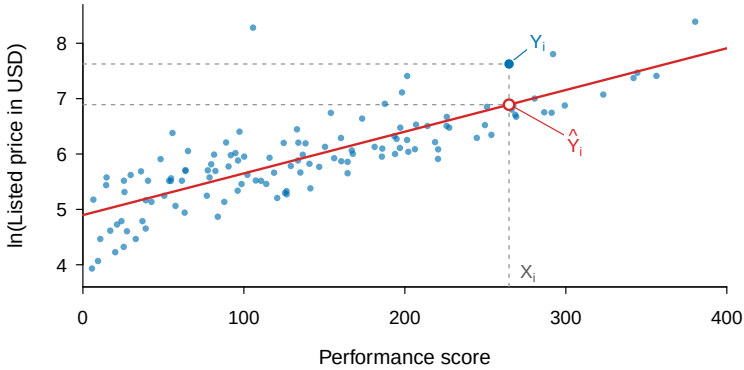
- How does R choose the best-fitting line among all possible lines?
- How well does the line fit the observed prices, and how accurately can it predict prices for other video cards?
- These questions guide the rest of this chapter and the next

Model Fit and Least Squares

Fitted Values and Residuals

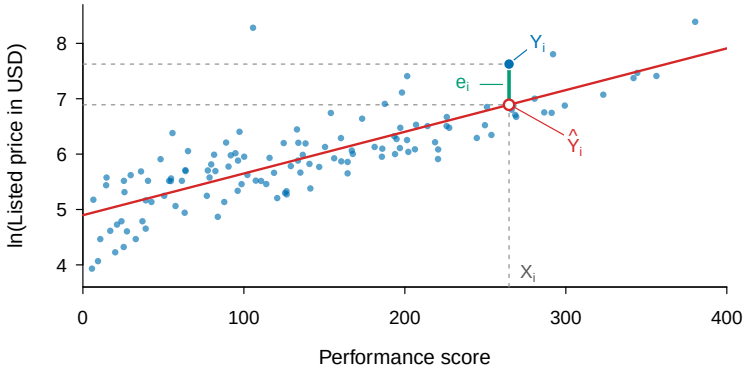
- **Data:** We observe the data recorded as the pairs (X_i, Y_i) for $i = 1, 2, \ldots, N$
- **Problem:** Choose a fitted line (b_0, b_1)
- **What to do:** A reasonable fitting rule minimizes differences between **fitted (predicted) values** and actual values across the sample
  - This gives us a measure of **in-sample model fit**: how well the fitted line explains the variation in the observed data
  - The smaller the deviations, the better the fit of the model to the sample
- The actual value minus the fitted value for a data point is called a **residual**
  - We often interpret residuals as the part of Y that cannot be explained by the fitted line

Fitted Values: Visualization



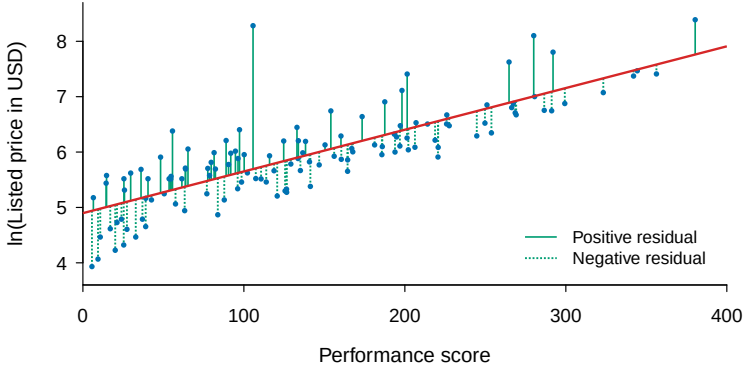
- The highlighted dot shows (X_i, Y_i) for GeForce RTX 3090
- The red line gives the fitted log price at each performance score: $\hat{Y}_i = b_0 + b_1 X_i$

Residuals: Visualization



- The residual is observed minus fitted: $e_i = Y_i - \hat{Y}_i$
- For the GeForce RTX 3090, $e_i = 7.63 - 6.89 = 0.74$ log-price units

Residuals: Visualization



- Each vertical segment joins an observed log price to its fitted value
- The fitted value and residual decompose Y_i into two parts:
  $Y_i = \hat{Y}_i + (Y_i - \hat{Y}_i) = \hat{Y}_i + e_i$
- $\hat{Y}_i$ is explained by the fitted line; e_i is the part left unexplained by this line

The Least Squares Criterion

- Our goal is to find a line that makes the residuals (prediction errors) as small as possible
- The most common approach is the **least squares** criterion
- **Definition:** Choose b_0 and b_1 to minimize the **sum of squared residuals**:

$$\sum_{i=1}^{N} e_i^2$$

  - It is also called “sum of squared errors” or “residual sum of squares”
- Why squared residuals?
  - Prevents positive and negative errors from canceling out
  - Penalizes larger errors more heavily
  - Provides a clear tradeoff: moving closer to some points often means moving farther from others
  - Many other reasons that will not be detailed in this course

Least Squares Solution

- Solving the **least squares** problem gives the following formulas:

$$b_1 = \frac{\sum_{i=1}^{N}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{N}(X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}$$

$$b_0 = \bar{Y} - b_1\bar{X}$$

  - s_{XY} is the sample covariance between X and Y
  - s_X^2 is the sample variance of X

Intuition behind the Least Squares Solutions

Intuition: The Intercept b_0

- With an intercept, the least squares **fitted line** passes through the point of **sample means** $(\bar{X},\ \bar{Y})$
- Proof: The least squares prediction at the average value of X ($\bar{X}$) is

$$\begin{aligned}\hat{Y} &= b_1\bar{X} + b_0 \\ &= b_1 \times \bar{X} + (\bar{Y} - b_1\bar{X}) = \bar{Y}\end{aligned}$$

  - The second equality plugs in the LS formula for the intercept

- The intercept b_0 adjusts the line vertically so it goes through the "center of gravity" of the sample $(\bar{X}, \bar{Y})$

Intuition: The Slope b_1

- Since $\bar{Y} = b_0 + b_1\bar{X}$ and the fitted line is $\hat{Y} = b_0 + b_1 X$, we obtain

$$\boxed{\hat{Y} - \bar{Y} = b_1(X - \bar{X})}$$

- b_1 measures the change in predicted Y ($\hat{Y}$) for each one-unit increase in X
- Here, X is performance score and Y is the natural log of listed price in USD

Quick Review: Correlation

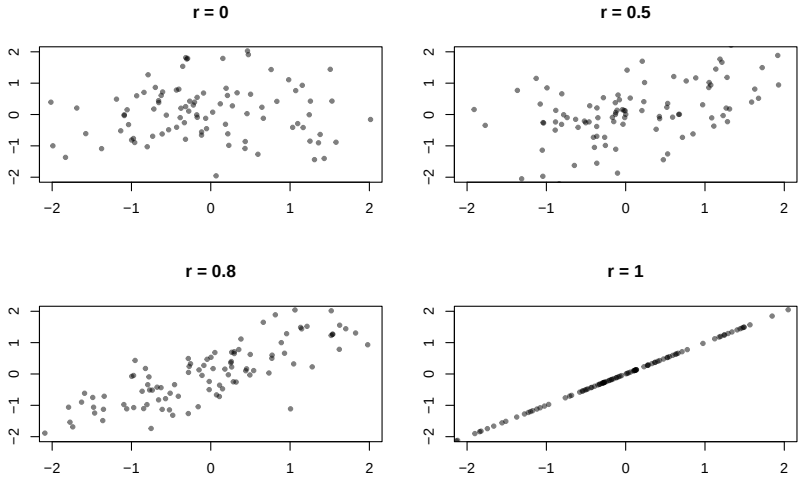
- The sample correlation coefficient between two variables is defined as:

$$r_{XY} = \frac{\sum_{i=1}^{N}(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{N}(X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{N}(Y_i - \bar{Y})^2}} = \frac{s_{XY}}{s_X s_Y}$$

- Notice that r_{XY} is a unitless measure of the direction and strength of the linear relationship between X and Y

Quick Review: Correlation

- Examples of samples with varying degrees of correlation



Intuition: The Slope b_1 and its Relationship with r

- The least squares slope captures the linear relationship between X and Y in the sample
- Each observed Y_i can be decomposed into a fitted value and a residual

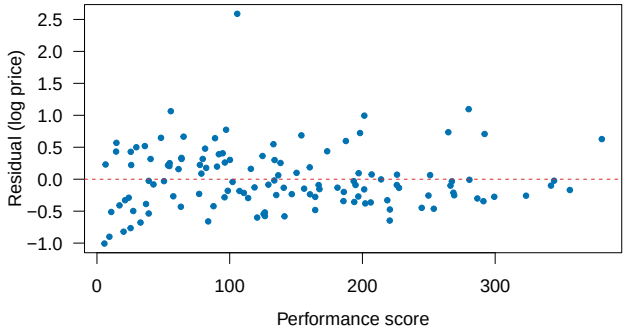
$$Y_i = \hat{Y}_i + e_i$$

- Under the least squares solution, the residuals have zero sample correlation with X: $\text{corr}(e, X) = 0$
- Zero correlation means no remaining linear relationship; a curved pattern may still remain

Intuition: The Slope b_1

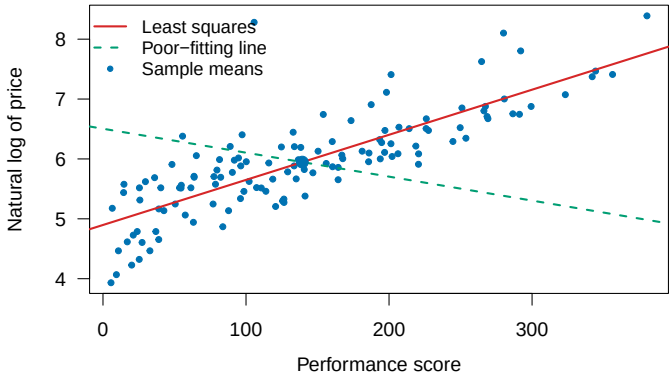
- In our 125-card sample, least squares with an intercept gives mean-zero residuals uncorrelated with performance score

```
round(mean(outlm$residuals), 10)
## [1] 0
round(cor(outlm$residuals, vc$performance), 10)
## [1] 0
```



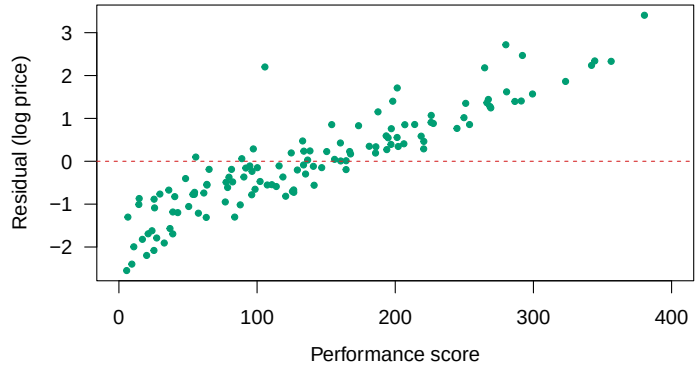
Intuition: The Slope b_1

- The dashed line passes through the sample means but has the wrong slope, so its residuals are correlated with X



Residuals under the Poor-Fitting Line

- Here, residuals are calculated using the poor-fitting line on the previous slide



- There is additional information in the residuals that can be captured by X
- **Question:** Suppose the poor-fitting line is $\hat{Y} = c_0 + c_1 X$. Given this scatter plot of residuals and X from the poor-fitting line, what's the relationship between b_1 from the least squared solution and c_1?

The Relationship between b_1 and r

- Why don't we just use r as the slope under the least squares solution?
- **Reason:** r_{XY} is unitless, but b_1 measures units of Y per unit of X
- b_1 can be written as the ratio of the sample covariance to the sample variance of X

$$b_1 = \frac{s_{XY}}{s_X^2} = \frac{s_{XY}}{s_X \cdot s_Y} \cdot \frac{s_Y}{s_X} = r_{XY} \cdot \frac{s_Y}{s_X}$$

Intuition behind the Least Squares Solutions: Summary

For least squares with an intercept, in the fitted sample:

- The intercept b_0 ensures that the prediction at the sample mean of X is the sample mean of Y

$$X = \bar{X} \implies \hat{Y} = \bar{Y}$$

- The slope b_1 ensures that the residuals are uncorrelated with X

$$\text{corr}(Y - \hat{Y}, X) = \text{corr}(e, X) = 0$$

Interpretation of b_1: A Log-Transformed Outcome

- Suppose we fit the model $\widehat{\log(Y)} = b_0 + b_1 X$
- A one-unit increase in X is associated with a b_1-unit change in predicted $\log(Y)$
- On the original scale, this corresponds to an approximately $100b_1\%$ change in predicted Y
  - In our video card example, $b_1 \approx 0.0075$: a one-point increase in performance score is associated with an approximately 0.75% increase in predicted price
- **Why?** Let $\hat{Y}_0$ and $\hat{Y}_1$ be the predictions on the original scale at X_0 and $X_0 + 1$

$$\log(\hat{Y}_1) - \log(\hat{Y}_0) = b_1 \quad \Longrightarrow \quad \frac{\hat{Y}_1}{\hat{Y}_0} - 1 = e^{b_1} - 1 \approx b_1$$

  - The approximation works well when b_1 is not too large

Interpretation of b_1: Which Variables Are Logged?

- The interpretation depends on whether we take the log of X, Y, or both

Fitted model	Increase in X	Change in predicted Y
$\hat{Y} = b_0 + b_1 X$	One unit	b_1 units
$\widehat{\log(Y)} = b_0 + b_1 X$	One unit	Approximately $100 b_1$%
$\hat{Y} = b_0 + b_1 \log(X)$	1%	Approximately $b_1/100$ units
$\widehat{\log(Y)} = b_0 + b_1 \log(X)$	1%	Approximately b_1%

A Goodness of Fit Measure: R^2

Decomposing the Variance

- For least squares with an intercept, $Y_i = \hat{Y}_i + e_i$ gives this sample variance decomposition

$$\begin{aligned} \text{var}(Y) &= \text{var}(\hat{Y}) + \text{var}(e) + 2\text{cov}(\hat{Y}, e) \\ &= \text{var}(\hat{Y}) + \text{var}(e) \end{aligned}$$

  - Why is $\text{cov}(\hat{Y}, e) = 0$? Recall that $\hat{Y} = b_0 + b_1 X$ and residuals are uncorrelated with X

- This is equivalent to

$$\underbrace{\sum_{i=1}^{N}(Y_i - \bar{Y})^2}_{\substack{\text{Total Sum of Squares} \\ \text{SST}}} = \underbrace{\sum_{i=1}^{N}(\hat{Y}_i - \bar{Y})^2}_{\substack{\text{Regression Sum of Squares} \\ \text{SSR}}} + \underbrace{\sum_{i=1}^{N} e_i^2}_{\substack{\text{Error Sum of Squares} \\ \text{SSE}}}$$

A Goodness of Fit Measure: R^2

- The fitted line captures more of the sample variation when SSE is small relative to SST
  - Fit would be perfect if SSE $= 0$ (equivalently, SST $=$ SSR)
- R^2 measures goodness of fit for least squares with an intercept

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$$

- Two interpretations of R^2:
  - Proportion of sample variation in Y explained by the fitted line
  - Square of the sample correlation between X and Y in simple linear regression

R^2 in R

```
summary(outlm)$r.squared
## [1] 0.6644497
cor(vc$performance, log(vc$price_usd))^2
## [1] 0.6644497
```

- About 66.4% of the sample variation in natural log listed price is explained by the linear fit using performance score

Misuse of R^2

- R-squared is often misused
- "High" values of R^2 ($R^2 \geq 0.8$) are often associated with claims that the model is:
  - Highly accurate for prediction
  - Adequate or correctly specified, meaning "valid"
- A high R^2 alone guarantees neither accurate predictions for new observations nor a correctly specified model

Appendix

Appendix: Derivation of LS Slope

For a least squares line with an intercept, the sample covariance between e and X is zero. We can use this property to derive the formula for b_1.

Verification:

Since the sample mean of the residuals is zero,

$$\text{cov}(e, X) = \text{cov}(X, e) = \frac{1}{N-1}\sum(X_i - \bar{X})(e_i)$$

Substitute $e_i = Y_i - b_0 - b_1 X_i$:

$$\sum(X_i - \bar{X})(Y_i - b_0 - b_1 X_i) = \sum(X_i - \bar{X})(Y_i - [\bar{Y} - b_1\bar{X}] - b_1 X_i) = 0$$

where $b_0 = \bar{Y} - b_1\bar{X}$

Appendix: Derivation of LS Slope

Verification (continued):

$$\sum(X_i - \bar{X})(Y_i - [\bar{Y} - b_1\bar{X}] - b_1X_i) = 0$$

or

$$\sum(X_i - \bar{X})(Y_i - \bar{Y} - b_1(X_i - \bar{X})) = 0$$

or

$$\sum\left[(X_i - \bar{X})(Y_i - \bar{Y}) - b_1(X_i - \bar{X})^2\right] = 0$$

Solving for b_1:

$$b_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$$

Glossary of Symbols

- X = independent or explanatory variable
- Y = dependent variable
- b_0 = least squares estimate of intercept
- b_1 = least squares estimate of regression slope
- e_i = least squares residual
- $\hat{Y}$ = fitted value
- s = sample standard deviation (the subscript identifies the variable)
- s^2 = sample variance (the subscript identifies the variable)
- r = sample correlation coefficient
- SST = total sum of squares
- SSR = regression sum of squares
- SSE = error sum of squares
- R^2 = coefficient of determination, goodness of fit measure

Important Equations

- Fitted value:

$$\hat{Y}_i = b_0 + b_1 X_i$$

- Residual:

$$e_i = Y_i - \hat{Y}_i$$

- Least squares formula for slope:

$$b_1 = \frac{\sum_{i=1}^{N}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{N}(X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}$$

- Least squares formula for intercept:

$$b_0 = \bar{Y} - b_1 \bar{X}$$

- Least squares line passes through point of means:

$$\hat{Y} - \bar{Y} = b_1(X - \bar{X})$$

Important Equations

- Relationship between slope coefficient and correlation:

$$b_1 = \frac{s_{XY}}{s_X^2} = r_{XY} \times \frac{s_Y}{s_X}$$

- For a least squares line with an intercept:

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}$$

Glossary of R Commands

- `abline(c(intercept,slope))` Adds one line through the current plot with the defined intercept and slope
- `anova()` Computes analysis of variance (or deviance) tables for one or more fitted model objects
- `apply(matrix, margin, function)` Applies the function to the rows or columns of a matrix « advanced R »
- `cov(x, y), cor(x, y)` Compute the covariance or correlation of variables x and y
- `lm(Y ~ X)` Fits a linear model with outcome Y and predictor X

Glossary of R Commands

- `points(x, y)` Plots the points specified by vectors x and y
- `predict()` Generates predictions from a fitted model
- `unique(vector)` Finds the unique values in a vector, e.g. `unique(c(1,1,5,4)) = c(1,5,4)`. Can also be used to remove duplicate rows in a data frame, e.g. `dfUnique=unique(df)`
- `with(df, expression)` Evaluates the R expression using the variables in the data frame df « saves writing out `df$varname` »